Linear Regression

Assumption

(1) Linear Relationship → $Y = mX + C$ → independent and dependent must be linear

(2) Normality of Residual → QQ Plot

(3) Homoscedasticity → Constant Variance of error → white test

(4) No Autocorrelation

(5) Multicollinearity → VIF

(6) No outlier problem → ~~Independent of error term~~ Box Plot

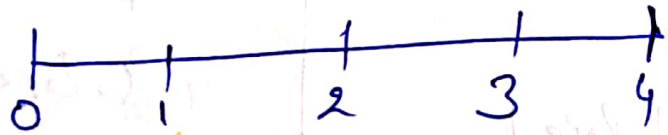
→ It states that error associated with one observation are not correlated with error of any other observation.

Error

→ Zero mean
→ Constant Variance → white test
→ Un Correlated → DW test
→ normally distributed → QQ plot

→ Linear relationship
→ Multicollinearity → VIF
→ No outlier problem

Durbin watson test (DW)



Variance Inflation factor (VIF)

$= \frac{1}{1 - R^2}$

VIF < 5

White test

$H_0 = \sigma_1^2 = \sigma^2 \Rightarrow$ Variance of error are equal
$H_1 = \sigma_1^2 \neq \sigma^2$

$p > 5\%$ → accept null hypothesis → Homoscedasticity

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OLS regression

Cost function $\Rightarrow \min \sum (Y_i - \hat{Y}_i)^2 = \min(SSE) \Rightarrow$ Least square criteria

Problem

↳ over fitting

↳ multi Collinearity

over fitting		under fitting	Generalized Model
Train	Acc = 90% ↓↓	60%	90%
Test	Acc = 70%	62%	89%
Low Bias High Variance		High Bias High Variance	Low Bias Low Variance

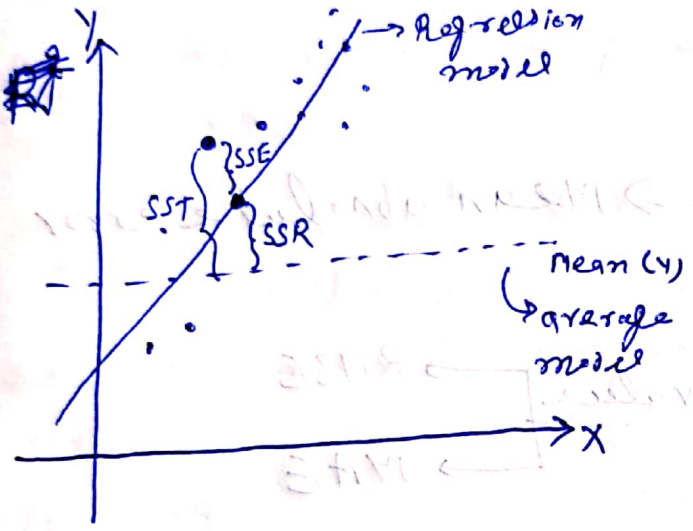
	L1 Regularization	
Ridge Regression (L2) Regularization	Lasso Regression (L1)	Elastic net Regression
~~Sum of squares~~ SSE + $\lambda \times \sum_{i=1}^{n} (slope)^2$	SSE + $\lambda \sum_{i=1}^{n} \lvert slope \rvert$	SSE + $\lambda_1 \sum_{i=1}^{n} (slope)^2$ + $\lambda_2 \times \sum_{i=1}^{n} \lvert slope \rvert$
Reduce Variance ↓ Reduce overfit	Reduce multicollinearity (feature selection)	Reduce overfit + Reduce multicollinearity (feature selection)

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Model Performance

(1) RSquare
(2) Adj RSquare
(3) Root mean Square error (RMSE)
(4) Mean absolute error (MAE)

(1) R^2



$$R^2 = \frac{\text{Sum of Square Regression}}{\text{Total Sum of Square}}$$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

$$SST = SSR + SSE$$

$$R^2 = 1 - \frac{\text{model error}}{\text{model2 error}}$$

(2) $R^2_{adj} = 1 - \frac{(1-R^2)(n-1)}{(n-k-1)}$

where,

n = number of sample

k = number of independent term in regression

It measure the proportion of variation by only those independent variable that really help in explaining the dependent variable. It penalizes for adding independent variable that doesn't help in predicting the dependent variable.

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(3) Root Mean Square Error (RMSE)

It explain how close the actual data points are to the model's predicted value. It measures standard deviation of the residual.

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}{n}}$$

$$MAE = \frac{\sum_{i=1}^{n} |(Y_i - \hat{Y}_i)|}{n} \Rightarrow \text{Mean absolute error}$$

Check deviation from Actual value → RMSE, MAE

for good model

high R^2_{adj}, low RMSE, low MAE

$\Rightarrow$ R^2 → Relative measure of fit

RMSE → Absolute measure of fit → R^2 v RMSE

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Regression result

	Coefficient	Standard Error	t Stat	P-value
Intercept	—	—	Coeff. / Standard Error	—
Independent Variable	—	—	—	—

t Stat = Coeff / Standard Error

ANOVA table

	df	Sum of Square SS	MS	F	Significance F
Regression	K	SSR	SSR/K	→ / ↑	—
Residual	n-k-1	SSE	SSE/n-k-1		—
Total	n-1	SST			

$$F = \frac{SSR/K}{SSE/n-k-1}$$

Cross Validation

Cross Validation is a resampling technique with the fundamental idea of splitting the data into 2 parts - training data and test data. Train data is used to train the model and unseen test data is used for prediction. If the model performs well over the test data and give good accuracy, it means the model hasn't overfitted the training data and can be used for prediction.

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k-fold cross validation

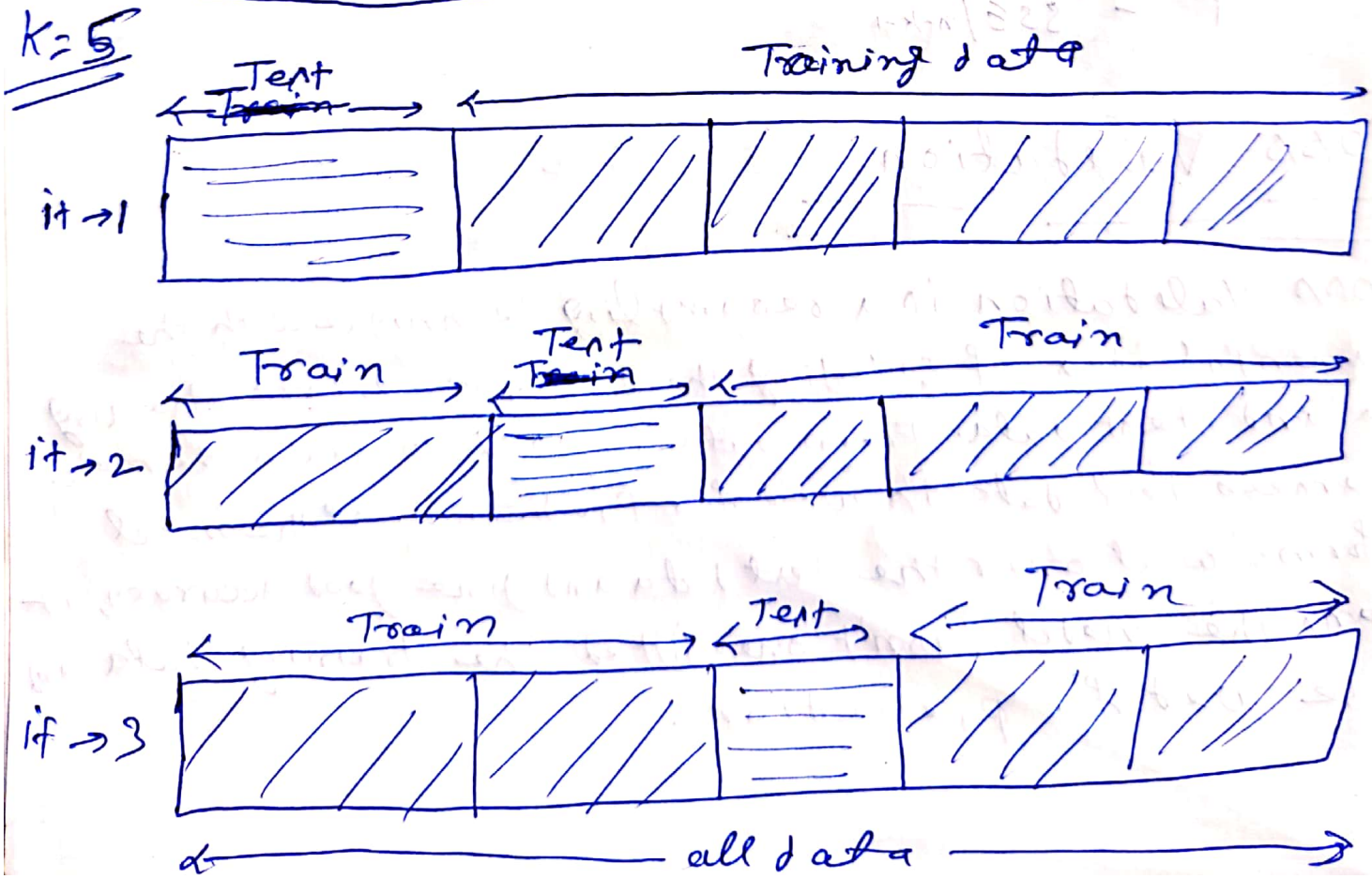
In this resampling technique, the whole data is divided into k sets of almost equal size. The first set is selected as the test and the model is traind on the remaining k-1 sets. the test error rate is then calculated after fitting the model to the test data.

In 2nd iteration, the 2nd set is selected as test set and remaining k-1 to train the data and error is calculated. This process is continues for all the k-sets

The mean error from all the iteration is calculated as the CV test error estimate.

$$CV = \frac{1}{K} \sum_{i=1}^{k} MSE_i$$

K=5



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Steps for Regression Model

1. Setup

(a) Create new enviroment in Python.

(b) Create requirement.txt

(c) Data transformation from input data

(d) Create Seprate folder for Requirement.txt, Exception handling, Loging, Data Load, Data Pipeline, Model training, Evaluation of model.

2. EDA

(a) Dataset information

(b) Data Validation

- → Check missing value
- → check Duplicate value
- → Check datatype
- → Check the number of unique values of each categorical column
- → Check Statistics of dataset
- → Check outlier → $Q1 - 1.5IQR < data < Q3 + 1.5IQR$
- → Check linear relationship b/w dependent with independent variable
- → Check Normality → Box Plot, QQ Plot, Histrogram, Kolmogorov Smirnov (KS) test, Shapiro wilk (W) test.

3. Feature Selection

(a) Correlation

(b) Forward/Backward/Stepwise based on cutoff AIC

4. Convert categorical variable to numeric → one Hot Encoding
5. Feature transformation → Normalize all data in one scale (Standard Scar)
6. Split data in Train and test
7. Train model with different model - OLS, Ridge, Lasso, Elasticnet, SVM, Xgboost, Adaboost, Random forest, decision tree

8. Evaluate Model Performance

1. R Square
2. Adj R Square
3. Root Mean Square Error (RMSE)
4. Mean Absolute error
5. Check regression assumption

9. Compare all model Performance and Select best performing model.
10. Create a Predict Column based on best model.

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